

PROJECT 08

AI Mock Interview & Assessment Platform

Master Architecture Specification & 25-Day Production Roadmap

Version	1.0
Date	September 2026
Team	10 Engineers / 2 Squads
Sprint	25 Days

Contents

- 1. Core Architectural Paradigm 3
- 2. Prioritized Product Scope 3
- 3. Production Tech Stack 4
- 4. Team Structure & Task Mapping 5
 - 4.1. Team A: Live Core Engine 5
 - 4.2. Team B: Platform & Exam Suite 5
- 5. 25-Day Sprint Timeline 6

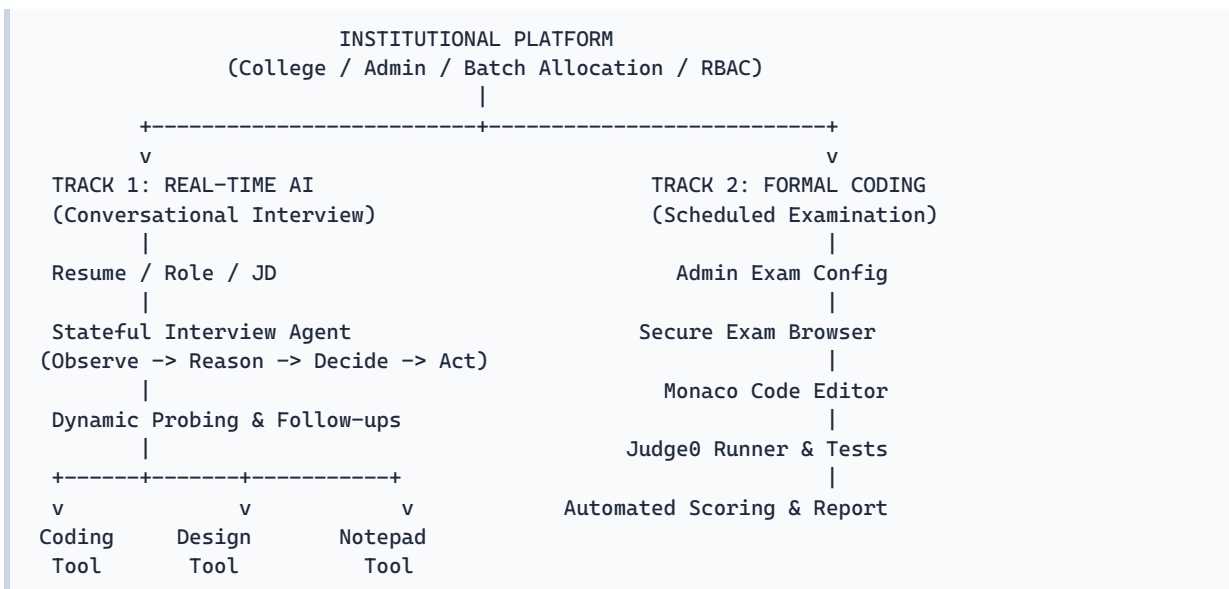
1. Core Architectural Paradigm

The platform is structured into two distinct assessment tracks managed under a unified institutional platform layer.

Track 1 (Real-Time AI) conducts conversational, adaptive interviews driven by a stateful agent loop. The agent dynamically invokes sub-tools — a coding workspace, system design canvas, or written notepad — as the interview progresses.

Track 2 (Formal Coding) provides standalone, scheduled coding examinations with a secure exam browser, sandboxed code execution, and automated scoring.

Both tracks share authentication and RBAC, a rubric scoring engine, and an institutional admin platform for college placement drives.



2. Prioritized Product Scope

Prior-ity	Module	Core Functionality	Track
P0	AI Real-Time Mock Interview	Resume/role/JD-based interview, dynamic fol- low-ups, adaptive difficulty, state/context man- agement, weak-area identification, session his- tory.	Track 1
P0	Resume Intelligence	PDF parsing, skill/project/experience extrac- tion, verifiable claim generation feeding the interview agent context.	Track 1
P0	Interview Agent	Stateful orchestration: Observe → Reason → Decide → Act → Verify.	Track 1

Prior-ity	Module	Core Functionality	Track
P0	Evaluation & Scoring Engine	Multi-dimensional scoring with transparent rubrics and session audit trail.	Shared
P1	Interactive Coding	1-on-1 coding tool invoked by the AI agent; candidate writes code, Judge0 executes, AI reviews logic & Big-O.	Track 1 Tool
P1	Standalone Coding Assessment	Formal scheduled exams, problem pool selection, test duration, candidate allocation, multi-language support.	Track 2
P1	Secure Exam (SEB)	Restricted/locked exam window, blur/tab-switch detection, unauthorized access limits.	Track 2
P2	System Design Assessment	AI-led architecture discussion (APIs, DBs, caching, failure modes) via dedicated workspace.	Track 1 Tool
P2	Written / Notepad	Free-form pseudocode, SQL queries, and architecture explanations without execution.	Track 1 Tool
P2	Voice Interview	Real-time STT/TTS, company-specific culture & interview simulations.	Track 1
P2	Competency Intelligence	Multi-session skill progression radar, skill-gap analysis, personalized 30-day plans.	Analytics
Inst	College / Admin Platform	Placement drive management, student cohort allocation, question pool authoring, faculty dashboard.	Admin Layer

3. Production Tech Stack

Layer	Technology	Responsibility
Backend API	Python (FastAPI) + Pydantic	Async REST, WebSocket pipelines, AI/ML integration.
Frontend	Next.js 14 + Tailwind + Lucide	SSR dashboards, client hydration for live cockpit & coding room.
Code Editor	Monaco Editor	VS Code engine in-browser: Python, Java, C++, JS; 300ms auto-save.
Code Execution	Judge0 CE Worker API	Isolated sandbox: 2.0s CPU timeout, 128MB RAM limit.
Database / ORM	PostgreSQL + SQLAlchemy + Alembic	ACID persistence for users, sessions, submissions, rubrics.

Layer	Technology	Responsibility
Real-Time Sync	FastAPI WebSockets + Redis	Bi-directional sync for interview state, timers, streaming probes.
AI / LLM	Gemini 1.5 Flash/Pro + OpenAI	Flash for live probing (<1s); Pro/GPT-4o for code review & coaching.
Exam Security	SEB / Browser Lock-down API	Fullscreen enforcement, tab-blur tracking, clipboard restriction.
Doc Processing	pdfplumber / pypdf	Deterministic resume parsing and claim extraction.
Testing / CI	Pytest, Playwright, GitHub Actions	Unit, contract, sandbox, and E2E tests on all PRs.

4. Team Structure & Task Mapping

4.1. Team A: Live Core Engine

A1. Monaco Editor & Workspace

- UI: Editor, language switcher, auto-save, stdin console.
- API: `POST /api/code/drafts`
- DB: `code_drafts` model.

A2. Sandbox & Test Runner

- UI: Execution console with test case results.
- API: `POST /api/code/execute` via Judge0.
- DB: `code_submissions` model.

A3. Live Cockpit & Session State

- UI: Cockpit with timer, round switcher, recorder.
- API: `/ws/session/{id}` WebSocket lifecycle.
- DB: `interview_sessions` model.

A4. AI Probing Agent

- UI: Thinking indicator, difficulty meter.
- API: `POST /api/ai/live-probe`
- DB: `agent_turns` model.

A5. Code Review & Big-O Engine

- UI: Big-O badges, code suggestions drawer.
- API: `POST /api/ai/code-review`
- DB: `code_reviews` model.

4.2. Team B: Platform & Exam Suite

B1. Auth, RBAC & Portal

- UI: Login, role management, onboarding wizard.
- API: `POST /api/auth/*` JWT + RBAC.
- DB: `users`, `student_profiles` tables.

B2. Resume Parser & Claims

- UI: Parsed resume viewer with skill pills.
- API: `POST /api/resumes/parse`
- DB: `resume_claims` model.

B3. Question & Problem Pool

- UI: Question bank, company filter, pool manager.
- API: `POST /api/questions/blueprint`
- DB: `question_bank`, `coding_problem_pool` tables.

B4. Exam Suite & SEB

- UI: Exam portal, countdown, SEB lockdown.
- API: `POST /api/exams/create`, `submit`, `session`.
- DB: `formal_exams`, `exam_allocations` tables.

B5. Rubrics, Analytics & Coach

- UI: Faculty dashboard, radar charts, PDF export.
- API: `POST /api/evaluations/score`, `coach/plan`.
- DB: `rubric_scores`, `coach_roadmaps` models.

5. 25-Day Sprint Timeline

Days	Phase	Milestones
1–3	Architecture	Monorepo initialized. PostgreSQL + Judge0 operational. Shared types frozen, schema models applied, mock APIs live.
4–10	P0 Verticals	All 10 members complete vertical slices with local tests. Resume Parser, Live Room, Monaco, Sandbox, and Auth functional.
11–15	Integration V1	Resume Upload → Blueprint → Cockpit → AI Probing end-to-end. Exam Portal connected to Judge0.
16–20	Integration V2	AI Agent invokes Coding Tool mid-interview → session → Rubric Scorer → Coach Plan export.
21–25	Hardening	Feature freeze Day 25. E2E tests, SEB validation, cloud deploy (Vercel + Render/AWS), viva prep.